

Article

Machine Learning in Creating Energy Consumption Model for UAV

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Abstract: The growing interest in the utilization of Unmanned Aerial Vehicles (UAVs) demands minimizing the costs of robot maintenance, where one of the main aspects relates to energy consumption. This manuscript presents a novel approach to create an energy consumption model for UAVs. The authors prove, based on experimentally collected data using a drone carrying various payloads, that Machine Learning (ML) algorithms allow to sufficiently accurately estimate a power signal. As opposed to the classical approach with mathematical modeling, the presented method does not require any knowledge about the drone's construction, thus making it a universal tool. Calculated metrics show the Decision Tree is the most suitable algorithm among eight different ML methods due to its high energy prediction accuracy of at least 97.5% and a short learning time which was equal to 2 ms for the largest dataset.

Keywords: energy consumption model; UAV; mobile robotics; machine learning



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1. Introduction

Over the last years, several papers related to Unmanned Aerial Vehicles (UAVs) have been published. Many of which investigate optimization techniques, especially in the package delivery task, proving system cost reduction [1–3] by utilizing drones that are more environmentally friendly [4–6] when compared to alternative solutions.

The popularity of investigating drones relates to their unlimited use cases. For example, the authors of [7] discussed UAV use cases in, among others, construction inspection, investigation and treatments in agriculture, transport and delivery applications, security supervising, and entertainment and filming. The authors argued that the energy consumption model should be based on the type of drone (whether it is a fixed-wing or a rotorcraft), altitude (propellers have to rotate faster at higher altitudes because of lower air density causing an increase in energy consumption), type of flight (hovering or forward flight), climbing speed, payload, and weather conditions (wind speed and direction).

Another interesting field of UAV usage relates to 5G technology. In [8], the authors utilized UAVs as flying base stations to improve transmission power. A similar solution is shown in [9] where drones carrying wireless network cards provide WiFi access to ground users. The authors of [10] extended functionalities by considering also the impact of renewable energy on creating a recharging schedule.

Zhang et al. in [11], based on a literature overview, summarized previous investigations and highlighted the fact that, in most papers, power estimation is based on simple, often linear, mathematical models, and there is little consideration of the nonlinear wind impact, which when omitted can cause the predicted energy to be even two times lower than the real consumption.

We can observe an increasing amount of papers proposing methods of minimizing UAV power consumption. The authors of [12] proposed optimizing the UAV flight time by using not only batteries as a power supply but also utilizing additional fuel cells and super-capacitors. In [13], the authors showed that public buses carrying recharge stations

can be used by surveillance UAVs in the city to extend the operational time. The importance of considering power aspects in UAV forces the researchers to be looking for convenient methods of UAV charging [14,15].

The reason for simplifying UAV's energy model is the high complexity of the task and the fact that an accurate modeling approach is not often necessary during consideration of drone use cases. Most mathematical models solve the power consumption forecast challenge by using linear models that can depend only on the drone's payload [4,16–19], or even by expressing the energy consumption as unit cost per traveled distance [1,5,20,21]. Some models consider the impact of flying altitude [22], especially for the surveillance use cases as a trade-off between consumed energy and a supervised area [23,24].

The linearity assumptions are valid for simplification when considering some use cases as optimal path planning for the fleet of UAVs delivering packages or in surveillance tasks using simulation. Nevertheless, the solution implementation can be infeasible in the real world. The existing nonlinearities can cause safety issues when power consumption is underestimated, resulting in an earlier battery's full discharge as shown in [25].

There are a few papers considering more input parameters such as a UAV payload and a varying velocity [26] in addition to the lift-to-drag ratio [27]. Only a fraction of methodologies utilize the wind impact on the drone due to uncommon wind measurement units, rarely available on commercial UAVs [3], and parameters describing the drone's rotors [28,29]. The authors of [30] also considered the power consumed by electronics, differentiating states when the drone is idle or armed. These models prove that the characteristic of power consumed in the function of total drone mass is nonlinear, similar to exponential. Moreover, considering the impact of wind, the power function is convex, with a local minimum. When the airspeed is low, the consumed power can be higher than in conditions of mild wind [11,26,28,29]. Unfortunately, the stronger wind has to be countered during flying causing an increase in energy demand. The main drawback of including information about wind in the energy model is the necessity of calibration using the anemometer's real data. This means there is a need of conducting tests in an unpredicted outside environment or in controllable form, using aerodynamic tunnels which can be unavailable for larger drones. However, the data from anemometers can be estimated based on the so-called wind triangle. The calculation is created by a flight vector v , a ground vector (from GPS (Global Positioning System)) w , and a wind vector u . When only w is given, v is estimated based on roll and pitch angles collected from the Inertial Measurement Unit (IMU) as shown in [31,32]. In [33], the authors proved that the Machine Learning (ML) algorithms can accurately estimate wind speed and direction utilizing this approach.

Concerning the present literature and investigations, we spotted the problem that researchers who do not want to examine deeply the relations between drone construction and its power consumption have to use simple analytical models which can be infeasible during real-world tests. We are aware that even the simplest models need to be properly identified so that the necessity of collecting power consumption data is unavoidable. For this reason, we verified available alternative methodologies for modeling energy consumption.

We decided to consider energy as a result of integrating the power signal. In [34], the authors claimed that the Artificial Neural Network (ANN) can outperform classical statistical forecasting approaches when a sufficient amount of data is available and can be as good when less data are delivered. On the other hand, the authors of [35], comparing 28 studies where ANN and regression models were used, asserted that ANN outperforms regression in 36%, outperformed in 14%, and had similar results in 50%. Nevertheless, the results show that both approaches can be used as they have similar performance, especially for large datasets. We chose to get a wider perspective and instead of using only ANN, we inspected different ML algorithms.

A justified usage of ML for energy forecasting is well described for smart buildings. In [36], the authors used three machine learning methods: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Artificial Neural Network (ANN) to predict the

energy consumption in smart buildings in Malaysia. The results show the SVN had the highest accuracy, but the training lasted 18 h. On the contrary, the training of KNN lasted only 40 s and the results were not significantly worse. Both methods outperformed the ANN. On the other hand, in [37], the authors used linear regression, SVM, and different configurations of multi-layer ANN to predict building energy consumption in Greece. The researchers obtained the smallest energy cost prediction error for the ANN method. Unfortunately, the performance of ANN is highly correlated with the size and quality of collected data. In [38], the authors predicted the building energy consumption based on the time of the day using ANN; the final error oscillated between 2% and 36% which makes it unpredictable. In [39], the authors examined 12 ML methods for predicting water and energy consumption in buildings again proving that is possible to acquire highly precise forecasts using artificial intelligence.

In the robotics field, there are still gaps in utilizing ML for energy consumption estimation. We already proved in [40] that the ANN can be utilized exchangeably with complex, analytical models for energy prediction of Unmanned Ground Vehicles. In this paper, we propose substituting a default mathematical modeling with a more universal ML algorithm to predict a drone's power consumption. The utilization of our approach is easier and faster because it does not demand knowledge about UAV construction. This is a significant advantage, especially for multi-robot system owners which do not have the resources for creating an analytical energy model for each robot. To prove the high performance of our original approach, we decided to collect power consumption data using a drone carrying different payloads in a real environment. Additionally, we investigated how choosing an ML algorithm impacts the final solution. We decided to compare the performance of eight algorithms by calculating the Root Mean Squared Error (RMSE) of the forecast power signal, the relative error of predicted energy, and the computational effort by measuring learning time.

The paper is divided as follows. Section 2 contains a description of test equipment and methodology. In Section 3, the research results are presented. Sections 4 and 5 show a discussion of the outcomes and conclusions of conducted research with directions for developing our research, respectively.

2. Materials and Methods

In this paper, we investigate energy consumption algorithms based on data gathered using one of our hexacopter presented in Figure 1.

The drone has six 18-inch propellers, weighs 6 kg, and has a Maximum Takeoff Mass (MTOM) of 13 kg. The custom-made construction has a carbon fiber frame and aluminum legs, with elements made in the Selective Laser Sintering process.

The power data were collected using a voltage and current Hall sensor Mauch HS-200-LV/HV capable to measure a current up to 200 A with 10 Hz frequency. To decrease the noise impact, each of ten consecutive power samples was averaged, resulting in the frequency reduction to 1 Hz.

Due to rarely available equipment for wind speed and direction on the drone's board, we used an indirect method for wind estimation based on UAV's orientation. Although, the approach described in the literature requires calibration. We were not interested in the designation of exact wind parameters but we want to know the impact on the power consumption; so, we assumed passing the drone's orientation as an input of the ML algorithm should be sufficient.

We collected data during automatic and more sophisticated manual mission execution by UAV without any and with additional payload weights of 2, 4, and 6 kg. The automatic measurements were collected for the hourglass shape trajectory with a goal velocity equal to 4 m/s. The path shape was chosen to check if the wind direction has a significant influence on the UAV flight. Additionally, during that shape, we turn in clockwise and counterclockwise directions equally; so, we could compare if the direction of turn is important. The hourglass-shaped path provides various training data. There are recorded

fragments with acceleration, deceleration, and constant speed flight. Moreover, the UAV is turning in both directions around the yaw axis. The purpose of the manual fly was to validate the training model during flying at different speeds.



Figure 1. The ROCON drone.

We gathered the following data:

- The altitude of the drone z_g (m);
- The linear drone's velocities (v_x, v_y, v_z) (m/s);
- The angular drone's velocities $(\omega_x, \omega_y, \omega_z)$ (rad/s);
- The orientation of the drone: roll, pitch, yaw (γ, θ, ψ) (rad);
- The total drone mass m (kg);
- The power consumption of the drone P (W).

We decided to calculate a correlation matrix to optimize the number of required inputs using the Pearson correlation coefficient. The approach allows sorting data importance and deciding which inputs can be reduced without significant influence on the algorithm accuracy.

The collected data were synchronized with power signal samples (1 Hz) and were divided into eight training and testing datasets, as shown in Figure 2.

1. We divided all of the data logs into two groups: collected during automatic and manual flights.
2. From the set containing automatic data logs, we have chosen a few data logs which had been shuffled. The rest of the logs from this set were merged with the other group containing manual test information.
3. The single training dataset contained from 10% up to 80% with a 10% step of the shuffled data logs.
4. The testing dataset contained the rest samples.

Each separate dataset was named as Division X%, which should be understood as X% of the samples from the colorful group (see Figure 2), is merged into the test data, and the rest is used as training data.

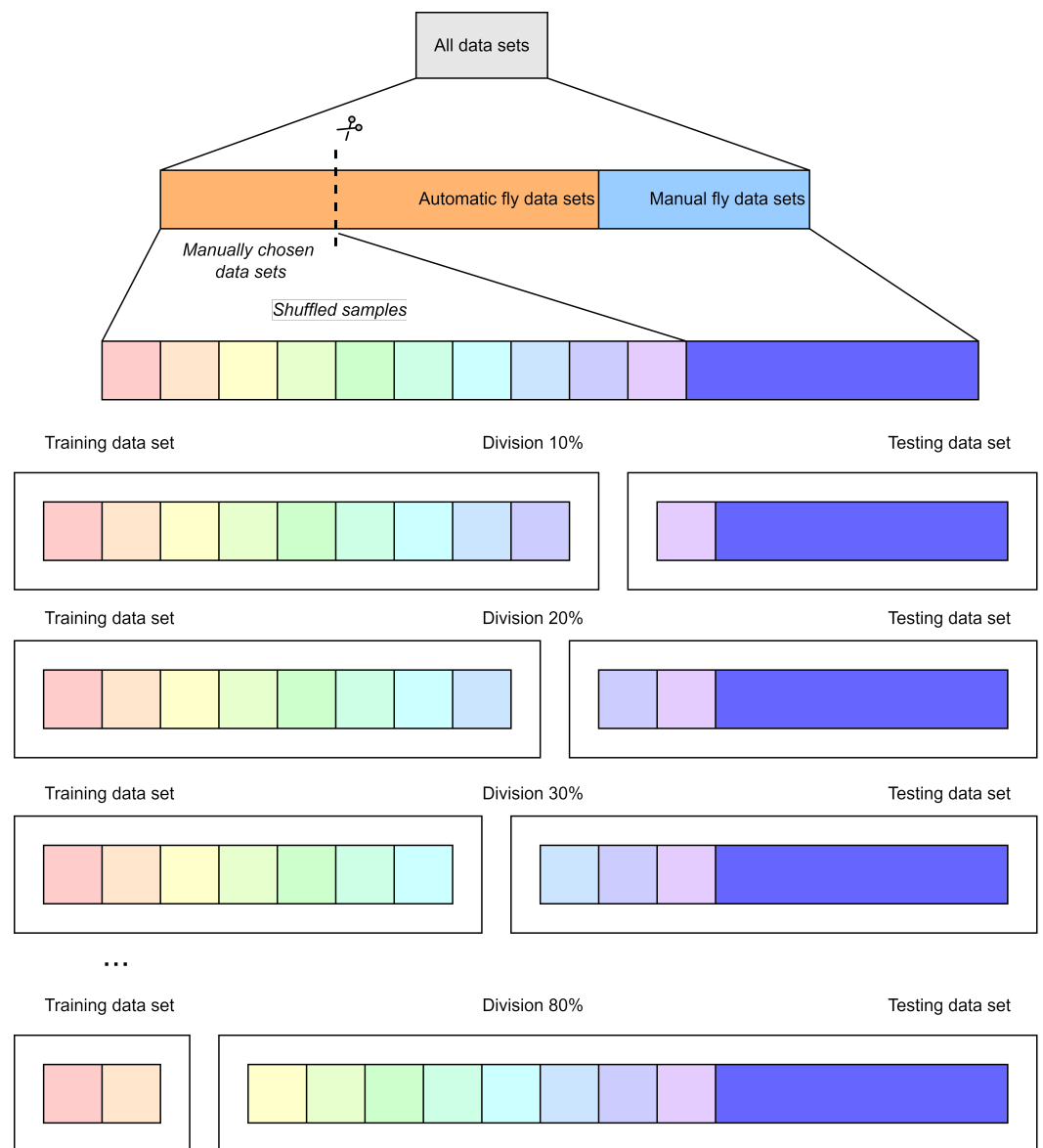


Figure 2. The dataset division procedure.

In our research, we used implementations available in scikit-learn Python's package. In the preprocessing, we scaled the collected data to unit variance and removed the mean. This provides the best performance when using methods requiring standard normally distributed data. To find the scaling formula z , we used (1):

$$z = \frac{x - u}{s}, \quad (1)$$

where x is the original value, u is the mean, and s is the standard deviation of the training dataset.

The purpose of this research is to prove that based on known velocities, altitude, orientation, and mass of the drone, a highly accurate energy consumption model can be established. We do not consider any information about UAV's construction such as the number of rotors, the parameters of motors and blades, etc. As a result, our solution would be compatible with any type of UAV.

In our research, we compared the following ML algorithms:

- Adaptive Boosting Decision Trees;
- Decision Tree;
- Gradient Boosting Decision Trees;
- K-Nearest Neighbors;
- Elastic Net Regression;
- Random Forest;
- Support Vector Machine;
- Feed-forward Artificial Neural Network.

To choose the best algorithm configuration, we utilized Grid Search Cross-Validation (CV) on the training dataset containing 50% of shuffled data logs. We were looking for the best solution among configurations created by mixing all of the parameters shown in Table 1. The approach allowed us to fully automatize the learning process. The best-found estimator's parameters are summarized in the next section.

Table 1. The set of configuration parameters considered during Grid Search CV. The bold and underlined parameters have been reported by Grid Search CV as the most suitable.

ML Method	Parameters
Adaptive Boosting Decision Trees	• The maximum number of estimators at which boosting is terminated: [30, 50, <u>100</u>, 1000]; • The loss function used for weight updating: [linear, <u>square</u>, exponential].
Decision Tree	• The function used for split quality measurement: [MSE, <u>Friedman</u>, MSE]; • The maximum depth of the tree: [2, 3, 4, 5, 6, <u>7</u>, 8]; • The maximum feature considered during split: [all <u>inputs</u> n, $\sqrt{n}$, $\log_2 n$]; • Maximum leaf nodes: [2, 3, 4, 5, 6, 7, 8, 9, <u>10</u>].
Gradient Boosting Decision Trees	• The maximum number of estimators at which boosting is terminated: [30, 50, <u>100</u>]; • The loss function used for weight updating: [least squares, least absolute deviation, <u>combination of two</u>, quantile regression]; • Learning rate: [0.01, <u>0.1</u>, 1, 10]; • The minimum number of samples required to split node: [<u>2</u>, 3, 5, 10]; • The maximum depth of a single estimator: [2, 3, <u>4</u>].
K-Nearest Neighbors	• The number of neighbors: [1, 3, <u>5</u>, 7, 11, 31, 99]; • The weight function: [all points weighted equally, <u>weights are proportional to the inverse of their distance</u>]; • The search algorithm: [<u>ball tree</u>, KD tree, brute force]; • The power metric: [<u>Manhattan distance</u>, Euclidean distance].
Elastic Net Regression	• The L1 and L2 regularization mixing parameter: [0, 0.01, 0.1, 0.5, <u>0.7</u>, 0.9, 0.95, 0.99, 1], where for 0 and 1 only L2 and L1 penalty is used, respectively; • The multiply penalty term factor: [<u>0.01</u>, 0.25, 0.5, 0.8, 1].
Random Forest	• The number of trees in the forest: [50, <u>99</u>, 120]; • The quality measure function: [<u>MAE</u>, MSE]; • The maximum feature considered during split: [all inputs n, $\sqrt{n}$, $\log_2 n$].
Support Vector Machine	• The kernel type: [<u>linear</u>, polynomial, radial basis function]; • The degree of polynomial: [<u>2</u>, 3, 4, 5]; • The regularization parameter: [0.001, 0.01, 0.1, 0.5, 0.8, <u>1</u>]; • The epsilon parameter: [<u>0.05</u>, 0.1, 0.2, 0.5].
Feed-forward Artificial Neural Network	• Hidden layers: [one with 5 neurons, one with 10 neurons, two with 5 neurons in each, <u>two with 10 neurons in each</u>]; • Activation functions: [the hyperbolic tangent, <u>the rectified linear unit</u>].

To assess algorithms' performance, we used the power signal's Root Mean Squared Error (RMSE) (2). We integrated the power signal using the trapezoidal rule to calculate real E and estimated energy $\hat{E}$ as (3) and (4), respectively. The relative error of predicted energy ϵ was calculated as (5). The average algorithm learning time is equal to a mean training time in Grid CV (only performed for the Division 50%) or a total learning time (in the rest of the Divisions).

$$RMSE = \sqrt{\frac{1}{n} \cdot \sum_{i=0}^{n-1} (p_i - \hat{p}_i)^2}, \quad (2)$$

$$E = \sum_{i=1}^n \frac{p_{i-1} + p_i}{2}, \quad (3)$$

$$\hat{E} = \sum_{i=1}^n \frac{\hat{p}_{i-1} + \hat{p}_i}{2}, \quad (4)$$

$$\epsilon = 100 \cdot \frac{|E - \hat{E}|}{E}, \quad (5)$$

where p_i is an i -th sample of the real power signal, $\hat{p}_i$ is an i -th sample of the predicted power signal, and n is the total number of samples in both power signals.

3. Results

We collected data during 9 separate automatic tests with different payloads (2×0 kg, 2×2 kg, 2×4 kg, and 3×6 kg). During each test, the UAV executed the mission until maximum battery discharge, resulting in up to 20 hourglass-shape paths with a 3 s hover between consecutive loops. An example path is shown in Figure 3.

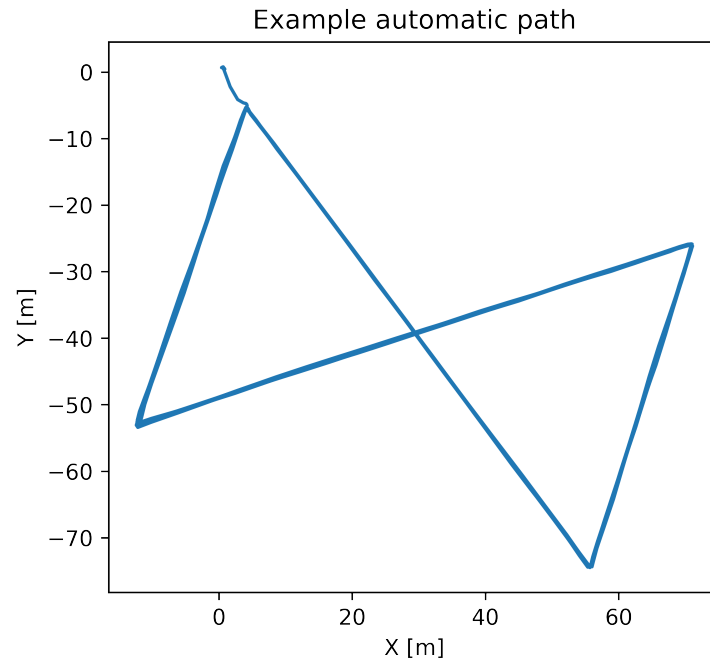


Figure 3. An example of UAV trajectory during the automatic test.

Additionally, we gathered sophisticated data during 3 manual tests with different payloads where the UAV velocity was randomly varied. An example of linear velocity along the drone's x-axis during the manual test is presented in Figure 4.

Using all collected datasets, we calculated correlations between collected data and we draw a heatmap presented in Figure 5. The relations between a power (output value) and the rest of the input values can be found in the last row of the heatmap.

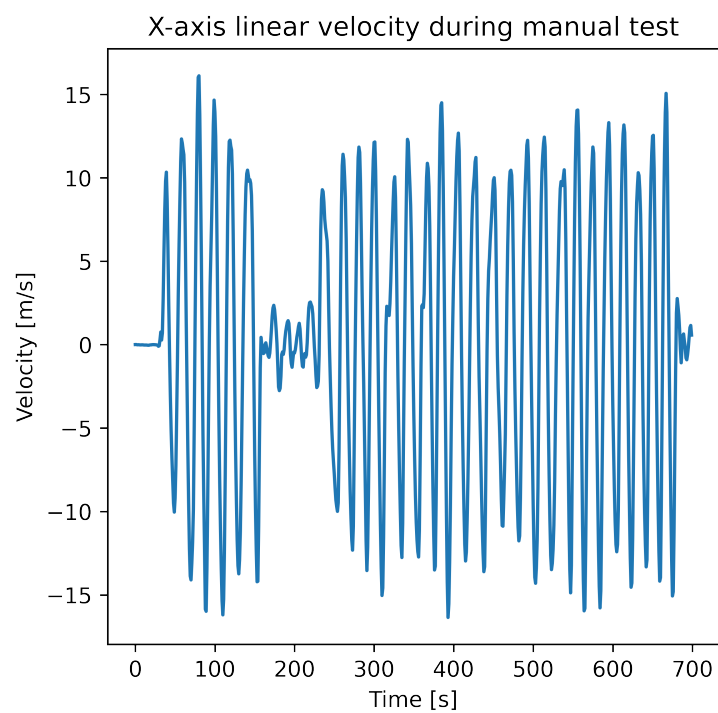


Figure 4. An example of linear velocity during a manual test.

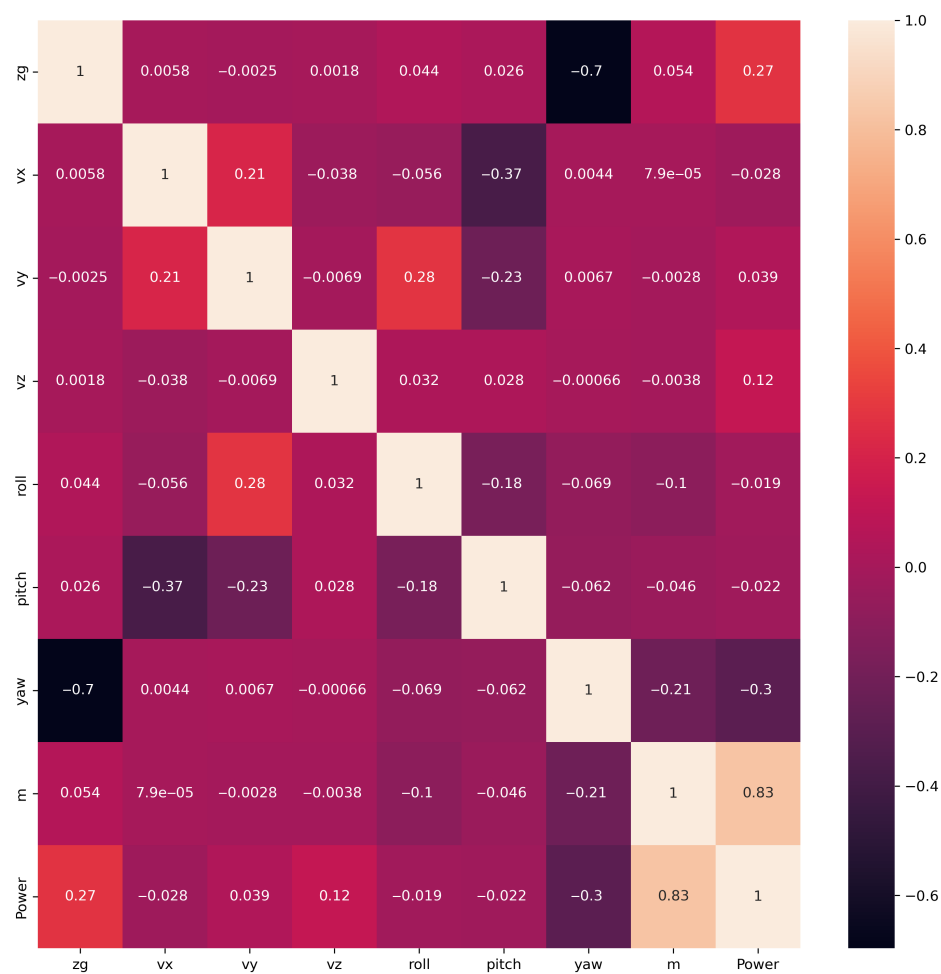


Figure 5. The correlation heatmap.

Due to the low relevance of angular velocities (w_x, w_y, w_z), we decided to remove them from further research to speed up the computation. The block diagram of the energy consumption model is shown in Figure 6.

The power samples for the complete dataset (training and testing) are shown in Figure 7 where colors correspond to the total mass in kg.

As part of the training set, we choose the four first data logs presented in Figure 7, the rest data logs are included in the testing dataset.

As Grid Search CV (for the Division 50%) results, we acquired the best parameters listed in bold in Table 1 for the investigated algorithms.

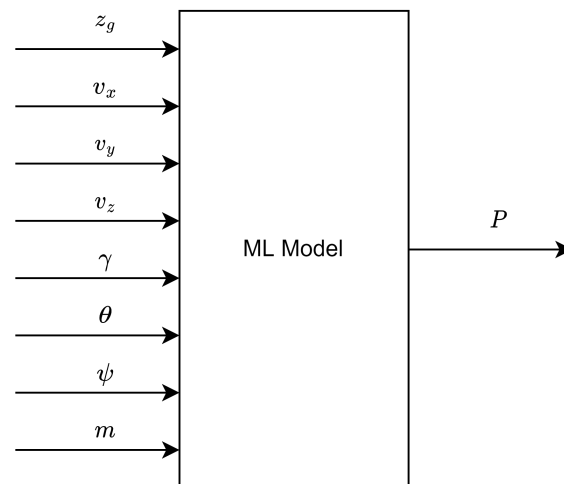


Figure 6. The inputs and output of ML model.

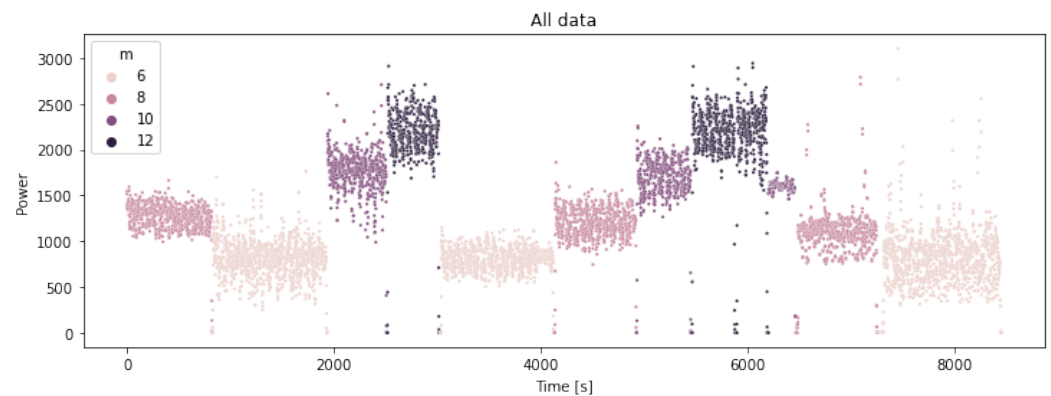


Figure 7. Completed dataset. Dataset contains 9 automatic logs with additional payloads (2 kg, 0 kg, 4 kg, 6 kg, 0 kg, 2 kg, 4 kg, 6 kg, 6 kg) and 3 manual logs with additional payloads (4 kg, 2 kg, 0 kg).

The outcome power signals for training and testing datasets for each of the considered ML method are shown in Figures 8–15.

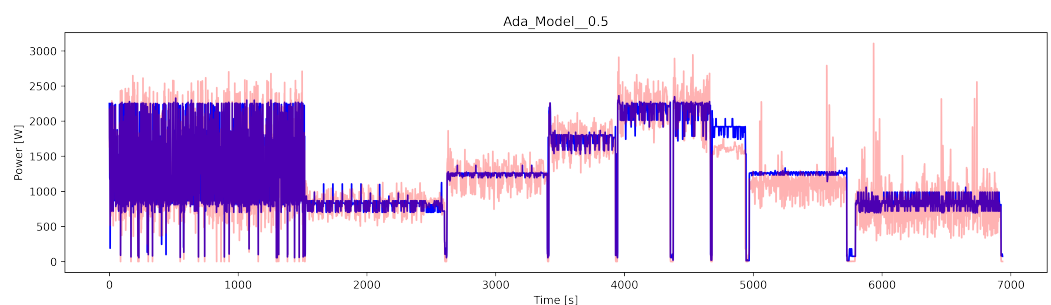


Figure 8. The real power (pink) and estimation (blue) generated by the Adaptive Boosting Decision Trees for the testing dataset.

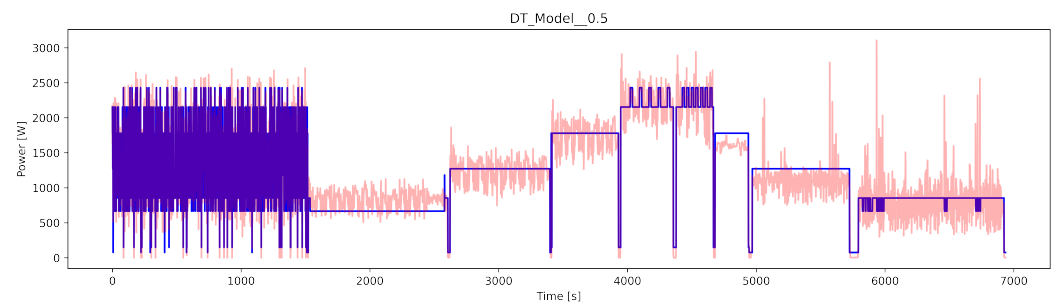


Figure 9. The real power (pink) and estimation (blue) generated by the Decision Tree for the testing dataset.

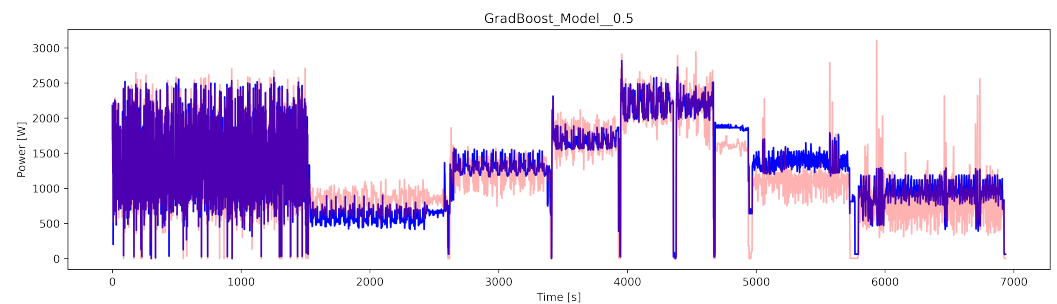


Figure 10. The real power (pink) and estimation (blue) generated by the Gradient Boosting Decision Trees for the testing dataset.

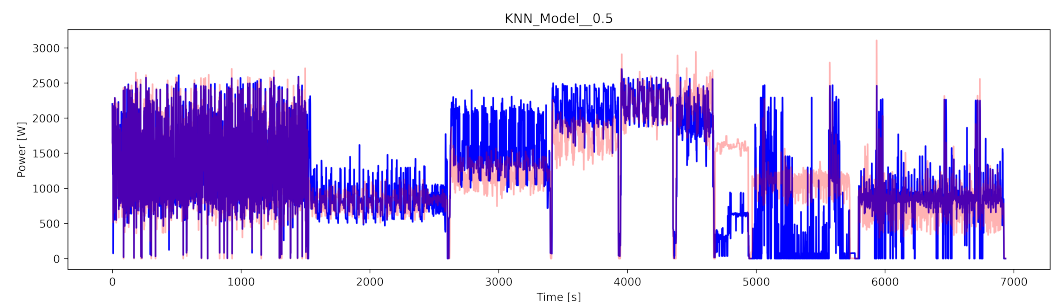


Figure 11. The real power (pink) and estimation (blue) generated by the K-Nearest Neighbors for the testing dataset.

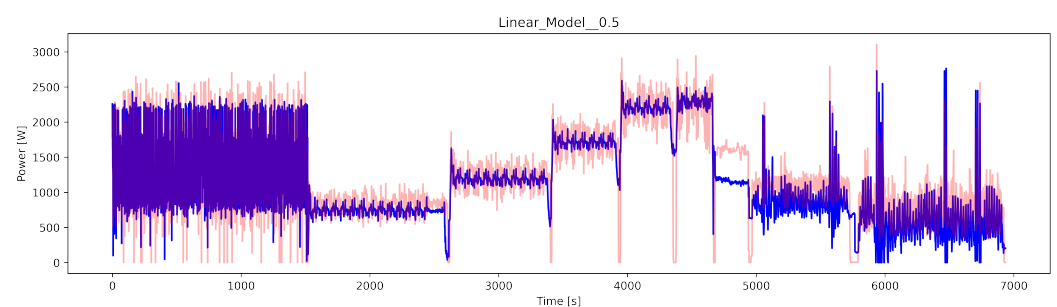


Figure 12. The real power (pink) and estimation (blue) generated by the Elastic Net Regression for the testing dataset.

The samples up to 1500 s are randomly chosen from the training dataset, as shown in Figure 2, the following samples contain data from testing logs.

Integration of ML algorithms output power signals produces the estimated consumed energy amount in Watt Hour (Wh), as shown in Table 2. The consumed energy calculated for the testing dataset based on measured power is equal to 2358 Wh.

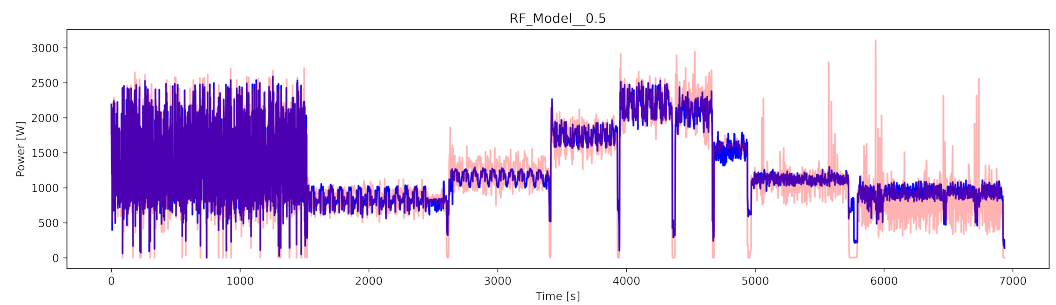


Figure 13. The real power (pink) and estimation (blue) generated by the Random Forest for the testing dataset.

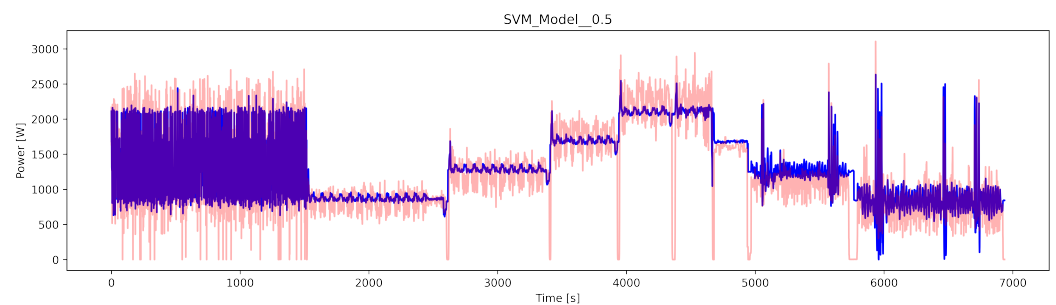


Figure 14. The real power (pink) and estimation (blue) generated by the Support Vector Machine for the testing dataset.

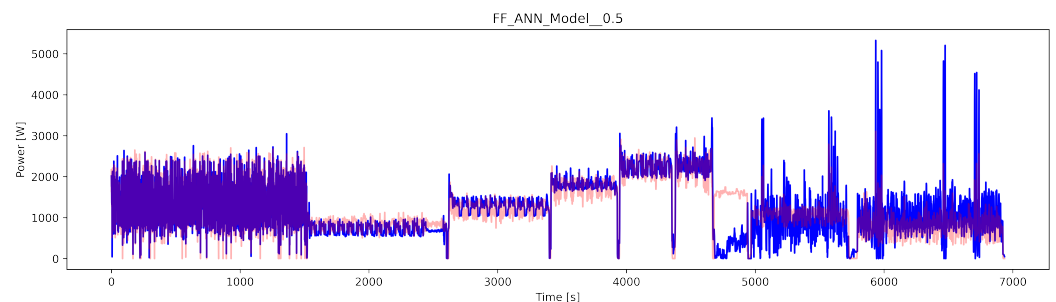


Figure 15. The real power (pink) and estimation (blue) generated by the Feed-forward Artificial Neural Network for the testing dataset.

Table 2. The relative energy prediction errors for the Division 50%.

ML Method	Estimated Energy [Wh]	Relative Error [%]
Adaptive Boosting Decision Trees	2422	2.7
Decision Tree	2391	1.4
Gradient Boosting Decision Trees	2424	2.8
K-Nearest Neighbors	2290	2.8
Elastic Net Regression	2196	6.8
Random Forest	2445	3.7
Support Vector Machine	2487	5.5
Feed-forward Artificial Neural Network	2375	0.7

In further research, we compared the relative error of estimated energy for all Divisions. The results are presented in Figure 16.

The RMSE for each Division is shown in Figure 17.

To compare computational efforts, we decided to calculate the average training time for each of the considered algorithms. The outcomes are presented in Figure 18.

Finally, we scored all of the Divisions allocating a maximum of 8 points and a minimum of 1 point for the best and the worst outcome in each category as follows:

- The maximum 8 points were allocated for the testing set with the lowest RMSE;
- The maximum 8 points were allocated for the testing set with the lowest relative error;
- The maximum 8 points were allocated for the testing set with the shortest learning time.

The results are presented in Table 3, and detailed outcomes of research are placed in Appendices for RMSE, the relative error of predicted energy, and average algorithm learning time in Tables A1–A3, respectively.

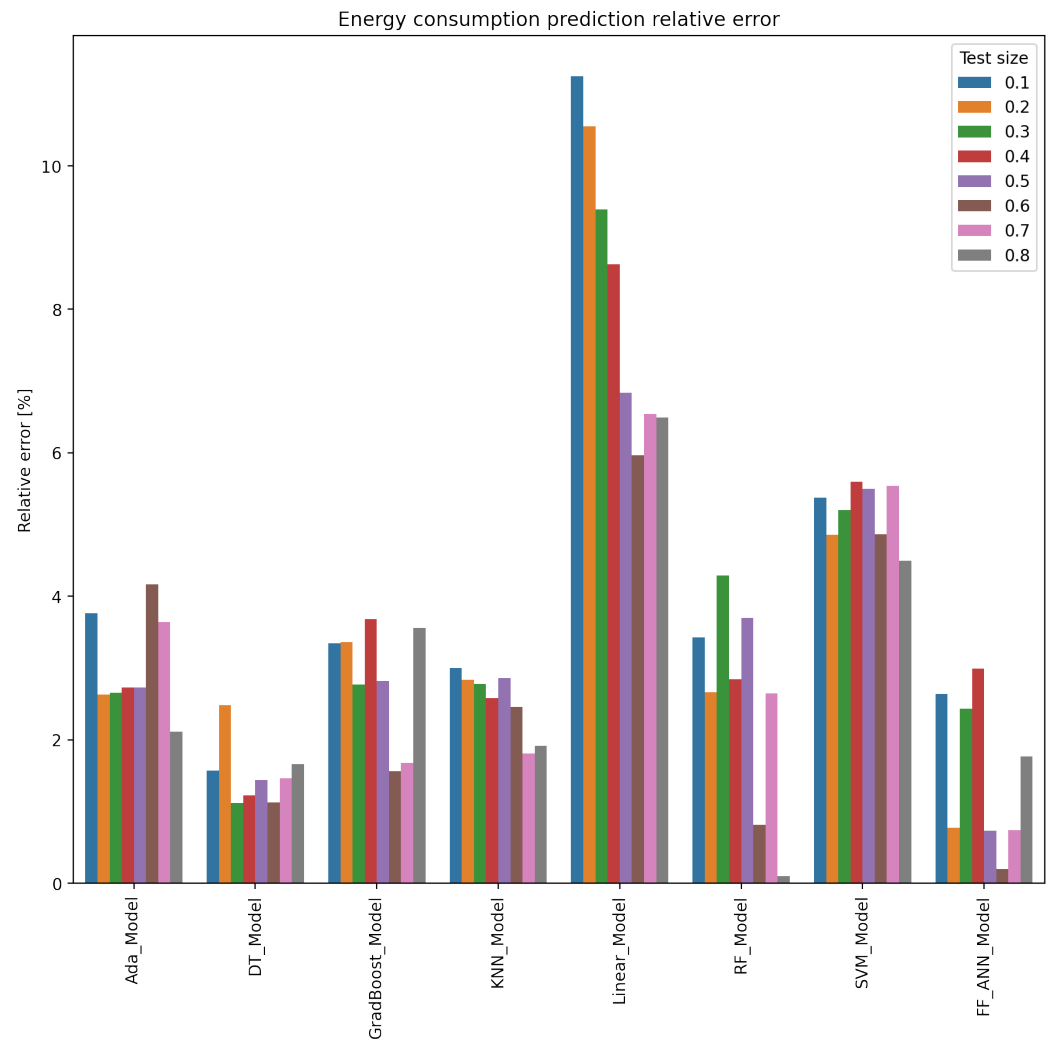


Figure 16. Comparison of relative error of predicted energy.

Table 3. The score table for the ML algorithm comparison.

Division [%]	10	20	30	40	50	60	70	80	Sum
Algorithm	Score								
Adaptive Boosting Decision Trees	14	15	17	17	15	13	11	15	117
Decision Tree	20	18	22	22	23	20	22	20	167
Gradient Boosting Decision Trees	15	13	14	12	14	15	16	14	113
K-Nearest Neighbors	14	12	13	15	13	13	14	15	109
Elastic Net Regression	12	12	12	12	13	13	13	13	100
Random Forest	13	15	12	13	12	14	14	16	109
Support Vector Machine	11	11	11	11	10	10	10	10	84
Feed-forward Artificial Neural Network	10	12	10	7	11	11	11	8	80

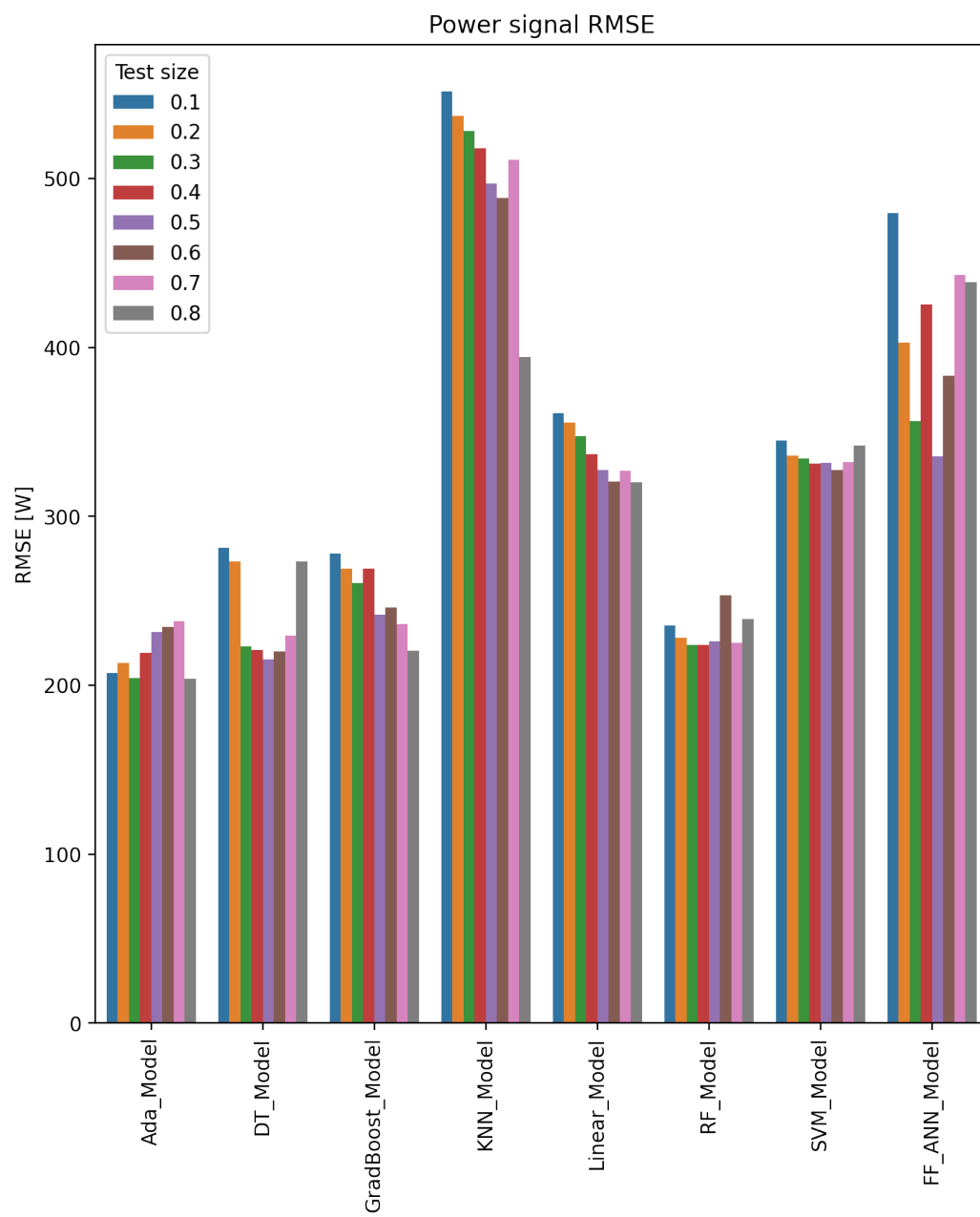


Figure 17. Comparison of RMSE value.

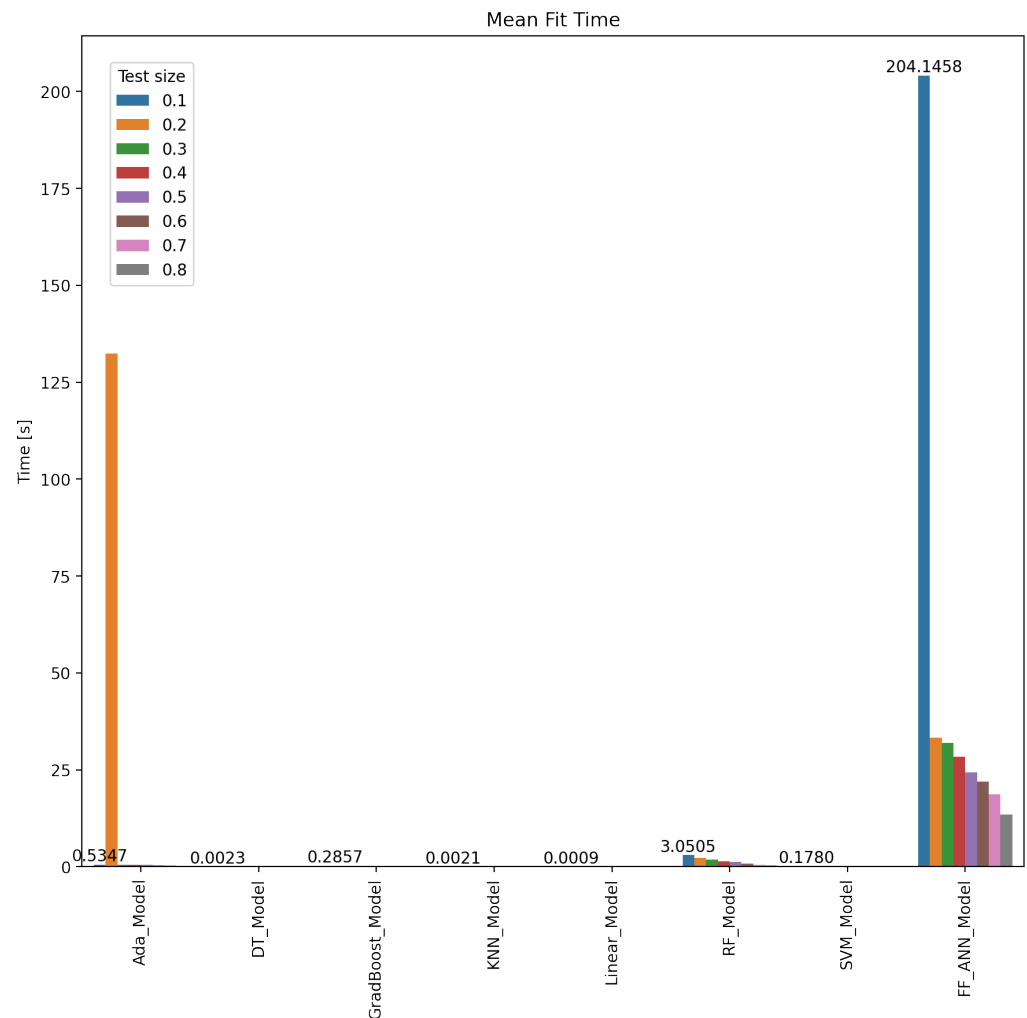


Figure 18. Comparison of average training time.

4. Discussion

Considering mathematically calculated correlations between consumed power and input parameters, it can be noticed that the most relevant information (correlation equals 0.83) is the total UAV mass. The impact could be also seen based on the collected power samples presented in Figure 7. There is a significant power offset between points corresponding to different weights of the robot.

The high correlation with the yaw angle (-0.3) results from directional wind impact during the tests. For a specific part of the executed trajectory, the yaw angle was constant but power consumption depended on the wind impact which created the correlation. We decided not to remove the yaw angle from model inputs because the difference between heading and flight speed is crucial when considering other flying machines such as airframes.

The next quite large correlation with the drone's altitude (0.27) relates to power peaks caused by altitude changes during manual test flights (see Figure 19). In the automatic approach, the drone was tested at the same altitude equal to 20 m. The impact of linear velocity along the drone's z-axis can be considered analogously.

The correlation of the rest of the inputs is relatively small, especially angular velocities which we omitted during the proceeding research.

Observing graphs presented in Figures 8–15, it can be noticed that for short periods, the power signal is changing. It indicates that all of the measured inputs have an impact on the power profile. Nevertheless, in the long term, the fluctuations could be replaced by a constant average power value calculated for a specific total drone's mass. Then, the

average power for a specific mass can be calculated and used to estimate an energy cost. It explains why many researchers decide to simplify the UAV model as a function of mass.

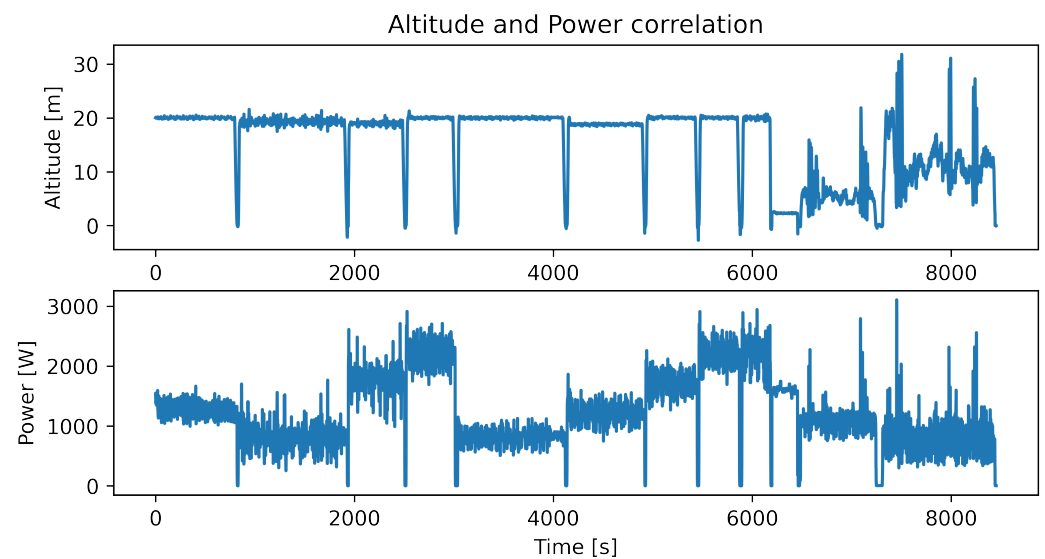


Figure 19. The consumed power for all collected data.

Considering the power measurements collected during manual flights, most of the tested methods were reliable. Unfortunately, two of them, KNN and ANN, generated highly unpredictable values.

Investigating RMSE and relative energy prediction errors, we can see a decreasing trend for the simplest algorithms (KNN and Elastic Net) with a decrease in the training set. It highlights the problem during training, causing a loss of approximating abilities of the algorithms. It results in a poor estimation of the testing dataset. Nevertheless, the KNN and Elastic Net algorithms are computationally light approaches.

The Artificial Neural Network is characterized by a low relative energy error, although it contains high RMSE. The ANN is good for predicting learned patterns but generates a high noise signal for the manual part of testing data (Figure 15) which makes this method uncertain. Additionally, it demands a lot of computation time, which increases for larger datasets.

More stable, but still not the best results are produced by the SVM algorithm. The training process is quite fast due to the simple linear kernel chosen in the Grid Search CV process. The disadvantage of the chosen SVM configuration is a linearization of the power model.

The best outcomes are generated by decision-tree-based methods. The learning time can vary depending on the size of creating a set of trees, for a single tree, the time is almost equal to zero, but for ensemble methods, it can be longer, sometimes unreasonable long as shown for Adaptive Boosting in Figure 18 for a 0.2 test size. The predicted power profiles are stable, working as well for learning hourglass-shaped pattern as for sophisticated manual flights.

Sorting the algorithms based on the obtained results from the best to the worst for the UAV energy consumption prediction task, we obtain the following list:

1. Decision Tree;
2. Adaptive Boosting Decision Trees;
3. Gradient Boosting Decision Trees;
4. Random Forest;
5. K-Nearest Neighbors;
6. Elastic Net Regression;
7. Support Vector Machine;
8. Feed-forward Artificial Neural Network.

Referring to Table 2, most of the algorithms overestimated consumed energy (the predicted energy value was higher than the real). Unfortunately, KNN and Elastic Net underestimated it which can cause serious problems. Overestimation can only decrease the system's performance as the drone will not use the whole available energy. In case of underestimation, the battery can fully discharge earlier than assumed and as a result, the drone will fall with a high probability of system destruction.

5. Conclusions

The Machine Learning algorithms are capable of properly estimating UAV energy consumption. The chosen algorithm for a specific application should consider the CPU (Central Processing Unit) computational power, especially when computer resources are limited. Although more advanced algorithms such as ANN and SVM have poor performance where training data are not diverse enough. One advantage of Neural Networks is that they can be easily calibrated online, thus improving prediction performance.

The main reason for the best accuracy of the decision tree is a high impact of only one input, i.e., the robot's weight. It means that the power signal could be well estimated using just a simple "if-else" statement considering only the mass. Due to that fact, the mathematical model simplifications used in other papers could be valid for simple applications in a stable environment, where the robot's mass is not changing and wind conditions do not change rapidly.

The utilization of Machine Learning is not a perfect solution for solving energy prediction tasks. The risk of underestimating energy disqualifies the approach for standalone usage in safety-critical systems. Nevertheless, it still can be considered when combined with redundancy safety precautions monitoring the battery state of charge. There is a high probability that a complex mathematical model can outperform the introduced approach, but we propose a trade-off solution providing satisfactory results with minimal effort. Designers can easily delegate the problem of modeling dependencies to computers. This universality is one of the advantages of ML over the classical mathematical approach. The ML model treats a robot as an abstract object which moves in the three-dimensional space. It is not necessary to consider the drone's construction as the methodology allows for calibrating the model automatically. The approach could be used by researchers and commercial users who do not want to create complicated analytical models to receive a highly accurate solution. The competitive edge of ML can be highlighted when it will be utilized in a multi-robot system. For people who use many different robots, the fully automated process of creating energy models could be very convenient as it saves time and money.

It can be observed that a significant part of the total drone's cost is the power supply subsystem which contains expensive, relatively short-living batteries. Additionally, constantly growing energy prices enlarge the UAV's costs. We believe that there will be demand for easily integrable energy models in the near future and the proposed approach will not remain only an academic consideration.

As part of our future work, we are planning to integrate the presented research with our previous work to investigate the solution performance in real-world applications. We would like to utilize energy prediction algorithms for a multi-robot system where UAVs realize tasks considering battery capacity constraints and cooperate with Unmanned Ground Vehicles carrying a battery swap station. The purpose of such a system is to provide optimal and long-term mission performance. Thanks to the advantages of the ML-based approach, we can easily implement a unified software package. It will be executed on any type of drone, calibrating estimation model parameters in an automatic manner. The methodology should allow us to create a distributed system where each robot will be a node-providing energy prediction to the system's supervisor. In the next step, we will investigate how to find an optimal way to generate power consumption model inputs based on the mission plan.

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Abbreviations

The following abbreviations are used in this manuscript:

UAV	Unmanned Aerial Vehicle
ML	Machine Learning
ANN	Artificial Neural Network
IMU	Inertial Measurement Unit
CV	Cross-Validation
RMSE	Root Mean Squared Error
KNN	K-Nearest Neighbors
SVN	Support Vector Machine
GPS	Global Positioning System
CPU	Central Processing Unit
MTOM	Maximum Takeoff Mass

Appendix A

Table A1. The RMSE value for the test dataset.

Division [%]	10	20	30	40	50	60	70	80
Algorithm	RMSE [W]							
Adaptive Boosting Decision Trees	207	213	204	219	231	234	238	203
Decision Tree	281	273	223	221	215	220	229	273
Gradient Boosting Decision Trees	278	269	260	268	241	245	236	220
K-Nearest Neighbors	551	536	527	517	496	488	511	394
Elastic Net Regression	361	355	347	336	327	320	326	320
Random Forest	235	228	223	224	226	253	225	239
Support Vector Machine	345	335	334	331	331	327	332	342
Feed-forward Artificial Neural Network	479	402	356	425	335	383	442	438

Table A2. The energy prediction relative error value for the test dataset.

Division [%]	10	20	30	40	50	60	70	80
Algorithm	Relative Error of Predicted Energy [%]							
Adaptive Boosting Decision Trees	3.8	2.6	2.7	2.7	2.7	4.2	3.6	2.1
Decision Tree	1.6	2.5	1.1	1.2	1.4	1.1	1.5	1.7
Gradient Boosting Decision Trees	3.3	3.4	2.8	3.7	2.8	1.6	1.7	3.6
K-Nearest Neighbors	3	2.8	2.8	2.6	2.9	2.5	1.8	1.9
Elastic Net Regression	11.3	10.5	9.4	8.6	6.8	6	6.5	6.5
Random Forest	3.4	2.7	4.3	2.8	3.7	0.8	2.7	0.1
Support Vector Machine	5.4	4.9	5.2	5.6	5.5	4.9	5.5	4.5
Feed-forward Artificial Neural Network	2.6	0.8	2.4	3	0.7	0.2	0.7	1.8

Table A3. The average learning time.

Division [%]	10	20	30	40	50	60	70	80
Algorithm	Average Algorithm Learning Time [ms]							
Adaptive Boosting Decision Trees	534	132,329	460	456	441	295	297	252
Decision Tree	2	2	1	1	1	1	0	0
Gradient Boosting Decision Trees	285	243	219	194	194	145	121	97
K-Nearest Neighbors	2	1	1	1	1	0	0	0
Elastic Net Regression	0	0	0	0	1	0	0	0
Random Forest	3050	2302	1844	1457	1299	821	548	324
Support Vector Machine	178	130	101	74	61	33	19	8
Feed-forward Artificial Neural Network	204,145	33,265	31,978	28,421	24,394	21,910	18,743	13,539

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